

CO₂/CH₄ Competitive Adsorption in Deformable Coal Nanopores Using Sequential GCMC–MD Approach

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Abstract:

Enhanced coal bed methane recovery through CO₂ injection involves competitive adsorption in coal nanopores and matrix swelling that alters storage and transport properties. In this work, I will employ a sequential GCMC–MD approach to investigate CO₂/CH₄ competitive adsorption in deformable coal nanopores. This project will provide adsorption isotherms, swelling–adsorption correlations, and clear trends in CO₂/CH₄ selectivity under varying pore deformation.

Reference:

- [1] Yang, Q., Xue, J., Lin, H., & Jin, Z. (2024). *Two-way coupling dynamics of CH₄ adsorption and coal matrix deformation: Insights from hybrid GCMC/MD simulations.* [\[Link\]](#)
- [2] Yang, Q., Xue, J., Lin, H., & Jin, Z. (2025). *Role of Adsorption-Induced Deformation on Gas Self-Diffusivity in a Flexible Microporous Coal Matrix.* *Langmuir.* [\[Link\]](#)
- [3] Zhou, W., Wang, H., Zhang, Z., Chen, H., & Liu, X. (2019). *Molecular simulation of CO₂/CH₄/H₂O competitive adsorption and diffusion in brown coal.* [\[Link\]](#)